LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Auto rotation, ground resonance, vertex ring, blade sailing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Auto rotation, ground resonance, vertex ring, blade sailing.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding rotor dynamics; Instructor supervision of theoretical explanation only.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the main rotor in translation flight?
(b) Define the term 'auto rotation' in the context of engine failure.
(c) List two common causes of excessive vibration in the rotor system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Auto rotation entry and control • Main rotor behavior during descent • Power management in auto rotation • Vibration checks and adjustments	L & Ds	PPT, IB	15 min
(b)	• Ground resonance mechanism • Vertex ring identification • Blade sailing characteristics • Safety implications of resonance	L & Ds	PPT, PDF	20 min
(c)	• Remedial actions for vibration • Monitoring device indications • Preventive maintenance checks • Reporting procedures for anomalies	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between ground resonance and blade sailing.
(b) Describe the immediate pilot actions required upon entering auto rotation.
(c) How does the vertex ring assist in diagnosing rotor imbalance?

12. Home Assignment and Follow Up	Prepare a brief note on the remedial actions for gearbox vibrators as covered in the next lesson plan (LP 09) to bridge the gap between rotor dynamics and monitoring devices.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Vibration of rotor and gearbox vibrators, disc cause effects on helicopter monitoring devices and remedial action. On hand experience to use vibration measuring equipment's.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min